Abstract

Carbon Nanotubes showing the expectational mechanical properties and enhancement of the other properties have been of much interest in creating some advanced Composites. They exhibit excellent tensile strength and elastic modulus, which is stiffer than steel and three to five times lighter. Products in many fields, composites reinforced CNTs are top of the line. Many concerns about global warming led to the sustainable production of products. In this paper, we have tried to make the CNT using Agriculture waste in a sustainable manner. We have used the way of burning the waste and using the chemicals that are commonly found in the laboratory. This process is repeated, and the sample quantity of CNT is increased. This sample is sent for testing and the results concluded that it is a Multi-walled Carbon nanotube. These CNT are used as a reinforcement in the preparation of Metal Matric Composites, the Matrix being Magnesium. This Metal Matrix Composite is produced using stir casting.

Introduction

The researcher's focus has been channelized towards the Carbon Nanotubes, since their discovery due to their exhibition of excellent properties. It has been identified as the "Material for the 21st Century".

Carbon Nanotubes (CNTs) are carbon rolled up into cylinders, due to their unique physics it has been used in much wide range of applications. They are known for their size, shape, and their large macromolecules. They can be easily explained as the sheet of graphite that is rolled into a cylinder. They can be in the fullerene structures of the carbon family. They got the name the Nano because of their diameter which is a few nanometers which are almost one-fifty thousandth width of the human hair.

Fig 1 CNT found in the powder form for commercial use.

The bonding nature between the nanotube was explained using orbital Hybridization from Quantum chemistry. Similar to the chemical bonding to that of Graphite which is sp^2 bonds. These nanotubes possess Van der Waals forces which allow them to naturally align themselves into ropes. They can transform some sp^2 bonds into sp^3 bonds, by merging under high pressure.

The properties of CNT in many various fields are remarkable which is explained in detail later in this paper. These properties make it most suitable for creating an advanced composite for the aerospace industry.

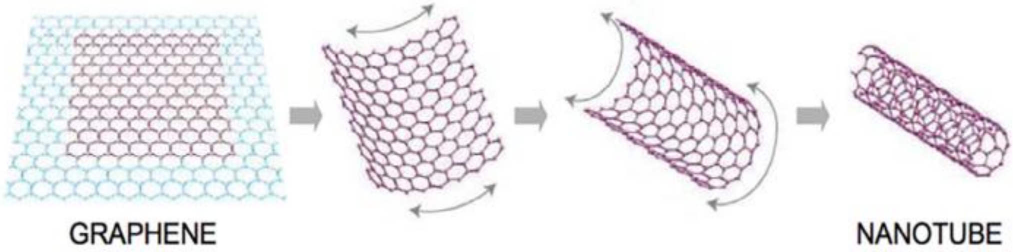

Fig. 1. Carbon nanotube.

Fig 2 Sheet of graphene rolling up to form carbon Nanotube.

There are two types of CNTs.

- Single Walled CNTs
- Multi-Walled CNTs

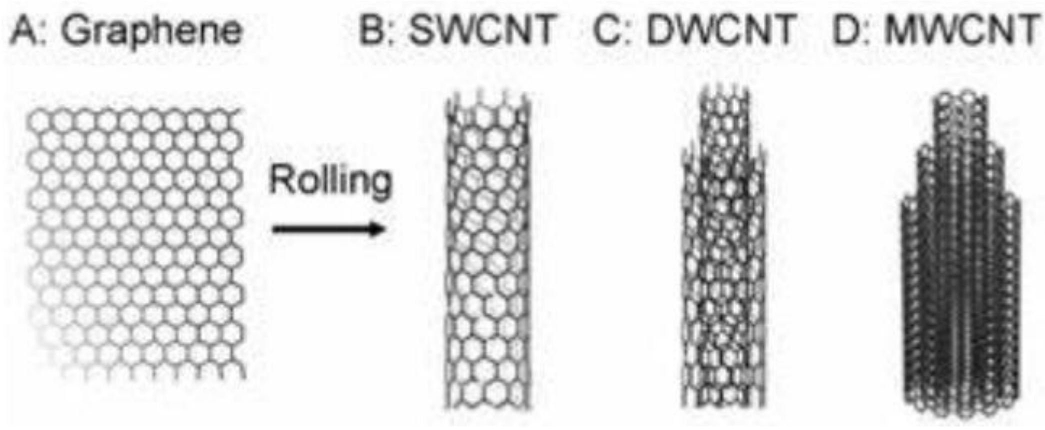

Fig 3 Representation of rolling of graphene into different types of CNTs

History

The discovery of the CNTs has been under discussion between researches for a long time. A large number of literature credits the discovery of the hallow, nanosized tubes to Sumio Iijima in 1991.

There was also a publication in the soviet Journal of Physical chemistry in 1952 by Radushkevich and Lukyanovich. This was left unnoticed because of the limitation of the publication being in the Russian language and also the Cold war.

There are many publications before 1991 which went unnoticed such as John Abrahamson presenting evidence of Carbon Nanotubes at the 14th Biennial Conference of Carbon at Penn State University and many more.

Iijima's discovery of carbon Nanotube created a lot of talks in the research community. Especially after his discovery of Carbon Nanotubes as the insoluble material of arc-burned graphite rods. This discovery has led to many independent discoveries and also different types of Carbon Nanotubes. This also led to many different scientists defining many different properties of the CNTs and their different applications.

Single Walled CNTs

The structure of the Single-Walled CNTs (SWNTs) can be visualized as graphene which is one-atom thick graphite in a cylinder. The diameter of SWNTs is close to 1 nanometer. There are different ways that the graphene can be molded into the cylinder which is represented by indices (n,m). These indices are named or called a Chiral vector.

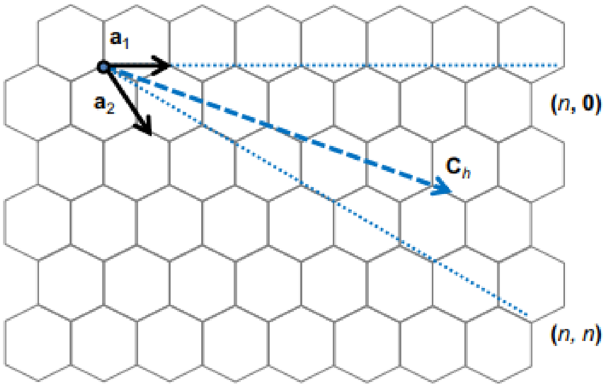

Fig 4 Chiral Vector representation on a graphene sheet of lattice.

The directions in the honeycomb crystal lattice structure of the graphene are represented by the unit vectors n and m. Depending on the values of n. m they are divided into

- Armchair (n=m)
- Zigzag (m=0)
- Chiral

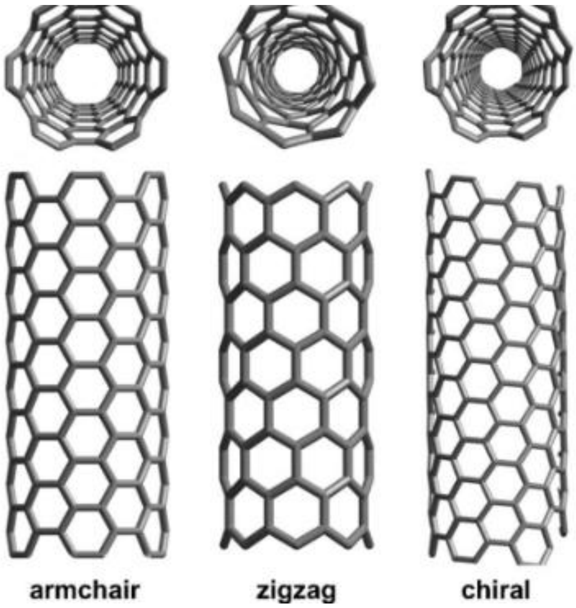

Fig 5 Different types of SWNTs

They are said to have superior electric properties than MWNTs. Because of their good conducting capabilities, they are the building blocks of the electric wire system. Single Walled Nanotubes are very expensive in the market of Carbon Nanotubes as there hasn't been any efficient and affordable synthesis method for their easy production.

Multi Walled Nanotubes

The structure of the MWNTs can be visualized as the multiple layers of graphene sheet rolled up to form a tubular shape. There are two main models in the MWNTs, they are

- Russian doll model: In this kind of model, the graphene cylinders are arranged in a concentric manner. Because it represents a Russian doll, it was given that name. An

example is small-diameter SWCNTs having larger-diameter SWNTs as the exterior nanotubes.
- Parchment model: In this model, a single sheet of graphene is rolled into multiple cylindrical rolls. The structure can be visualized to be a rolled-up newspaper or parchment.

The interlayer distance between two different graphene layers in multi-walled nanotubes is approximately $3.3A^0$(330 pm).

There is a particular type in the MWNTs that needs to be highlighted, they are DWNTs. They have the same properties as SWNTs but have an extra layer which is resistant to chemicals.

Properties

Strength:

CNTs compared to many metals such as steel and Kevlar have higher tensile strength. This strength is mainly attributed to sp^2 bonds between carbon atoms, compared to sp^3 bonds its stronger.

Other than being strong CNTs are also known for their elasticity. In ideally there is a limit for elasticity also but it only happens under very high pressure.

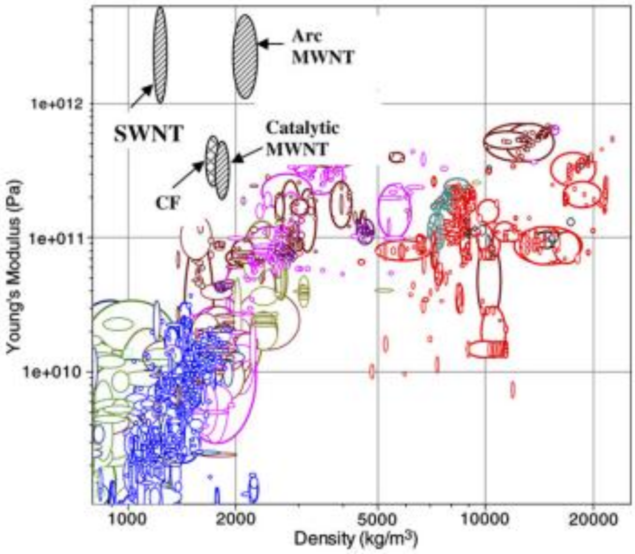

Fig 6 Modulus vs Density Chart depicting various engineering materials

From the above figure, we can observe that the CNTs can achieve to have high Youngs's Modulus for comparatively very low density.

Keeping the stiffness as importance in the strength. CNTs also show extremely good stiffness. In figure we can see that the high modulus to weight ratio there is a surge in price range as well.

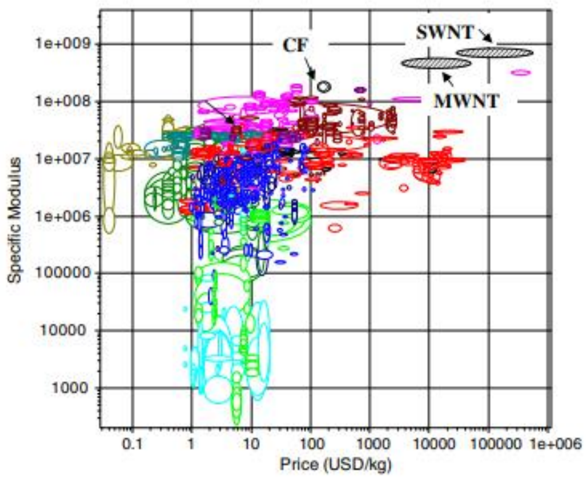

Fig 7 Chart between modulus to weight ratio and price

Comparison between different types of commercially available CNTs [9–13]

Supplier	Product	Purity (% nanotubes)	Elastic modulus (GPa)	Diameter (nm)	Density (g/cc)
MER	Catalytic MWCNT	>90%	200–500	140	0.1 Powder 1.9 bulk
MER	Catalytic MWCNT	>90%		35	1.9
MER	Arc MWCNT	35%	1000–3700	13–50	0.7 Powder 2.1 bulk
MER	SWCNT	12 wt% nanotube	1000+	1.2–1.4	0.025 Powder
Carbon solutions	AP-SWCNT	40–60%	1000+	1.4	1.4
Carbolex	AP-SWCNT	50–70%	1000+	1.4 (rope diam 20 nm)	1.4
Carbon solutions	RFP-SWCNT	60–80%	1000+	1.4	1.4
Carbon solutions	P2-SWCNT	70–90%	1000+	1.4	1.4
Carbolex	SE-SWCNT	70–90%	1000+	1.4	1.4
Thomas Swan	Elicarb high purity SWCNT	70–90%		<2 nm	

Table 1 comparing different types of commercially available CNTs

From the above table we can get a rough idea about the tensile strength and the compressive strength of the major types of CNTs such as SWNTs and MWNTs. The table gives detailed information about the commercially available CNTs and how purity, diameter, density and elastic modulus are interrelated.

Electrical Properties

The major effecting element on the electrical properties is the structure. It is that the structure should be in such a manner that the collisions among the electrons and atoms should be minimal, in such case the CNT is highly conductive. Due to the stronger bonds in the CHT they can handle more electric current than copper. The transportation of the electron which causes electricity only happens along the axis of the tube. CNTs also have a constant resistivity.

Thermal Properties

The strength of the bonds not only benefits the tensile strength but also in withstanding high temperatures. Due to these strong bonds, they are considered to be very good thermal conductors. Comparing them to a commonly used thermal conductor, copper wire, CNTs can transmit 15 times more. The effecting factors for CNT in thermal properties aretheathetemperature of the tubes and the environment around them.

Property	Unit	SWCNTs	MWCNTs
ρ	$g\,m^{-3}$	~ 1.3	~ 1.75
d_t	nm	0.6–0.8	5–50
AR	–	100–10,000	100–10,000
k	$W\,(mK)^{-1}$	3000–6000	3000–6000
ϱ	$\Omega\,m$	$1 \times 10^{-3} - 1 \times 10^{-4}$	$2 \times 10^{-3} - 1 \times 10^{-4}$
σ_{max}	GPa	50–500	10–600
$\bar{E}$	GPa	1500	1000

Table 2 comparison of different properties among different types of CNTs

From the above table, we can compare different types of CNTs such as SWCNTs and MWCNTs how they differ in different properties. The properties compared in the above table are Density, diameter, aspect ratio (AR), thermal conductivity (k), electrical resistivity, tensile strength, and young's modulus.

Physical properties of CNTs.

Properties	Values	Comparison
Mechanical	- Tensile strength:150 GPa - Young modulus (elasticity):1TPa - Low density	- 300 times more resistant and 6 times thinner than steel
Electrical	- Electrical resistivity: $10^{-4}\Omega$.cm - Current max density: 1013A.m^{-2} - Conductor or semiconductor	- Superior to copper* - Superior to that of classical conductor
Thermal	- Thermal conductivity ~3000 W/K.m	- Diamond ~1000 W/K.m

*Materials representing the highest values

Table 3 Comparing properties of CNT with commonly used material in particular sectors.

In this table major properties of the CNTs are compared with the commonly used material in that sector, from this we can see that the addition of CNT can be more advantages and create a more functionally advanced composite.

CNT in the Aerospace Sector

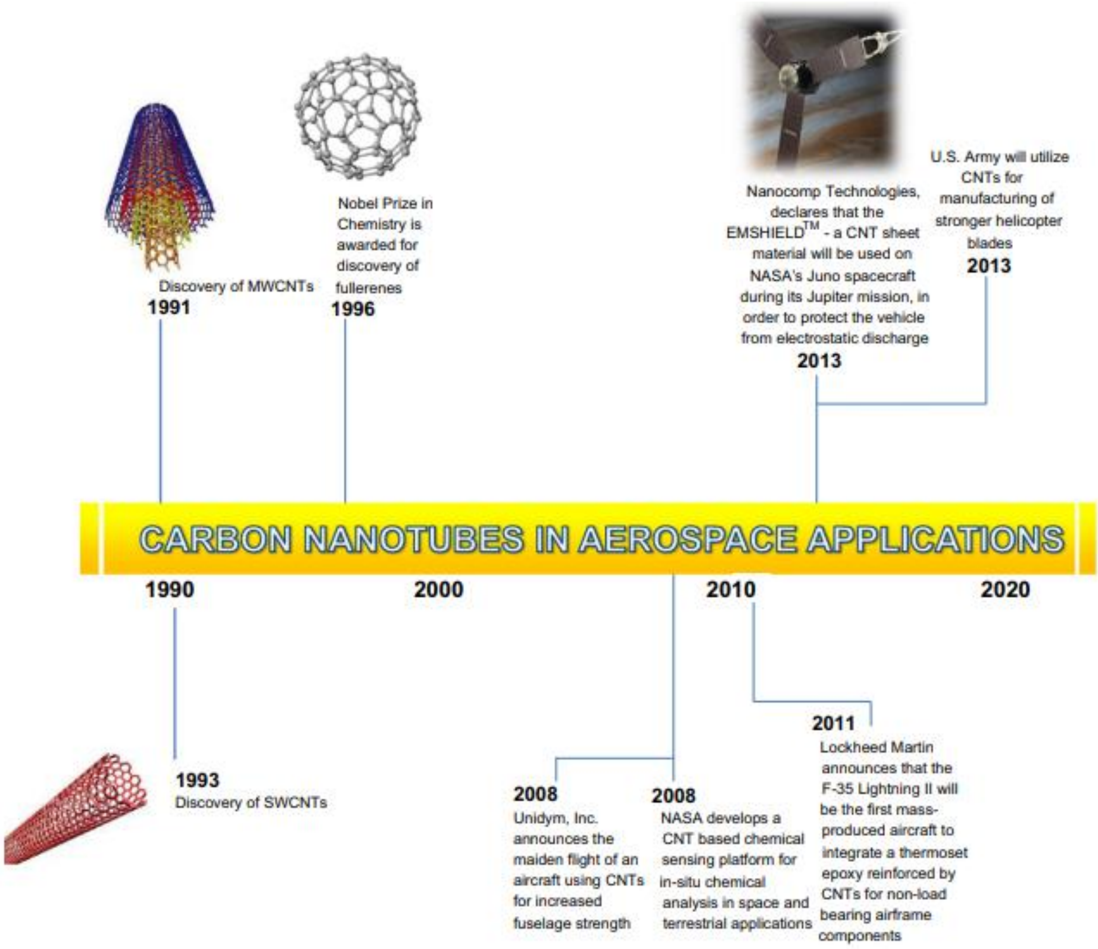

Fig 8 Representing different milestones of CNT in Aerospace Sector.

The potential advantages of using CNTs in aerospace applications, according to this blueprint of the figure given above, include lowered vehicle mass, greater functionality and endurance, higher damage tolerance, improved thermal protection, and control. CNTs have the potential to improve both the production and distribution of energy. Large-scale CNT production techniques, uniform CNT dispersion in composite materials, alignment and adhesion issues related to CNTs in reinforced polymers, a complete understanding of their toxicity, and the production of CNTs that are grain in volume and have a uniform size are some of the challenges that have been presented for using CNTs in aerospace applications.

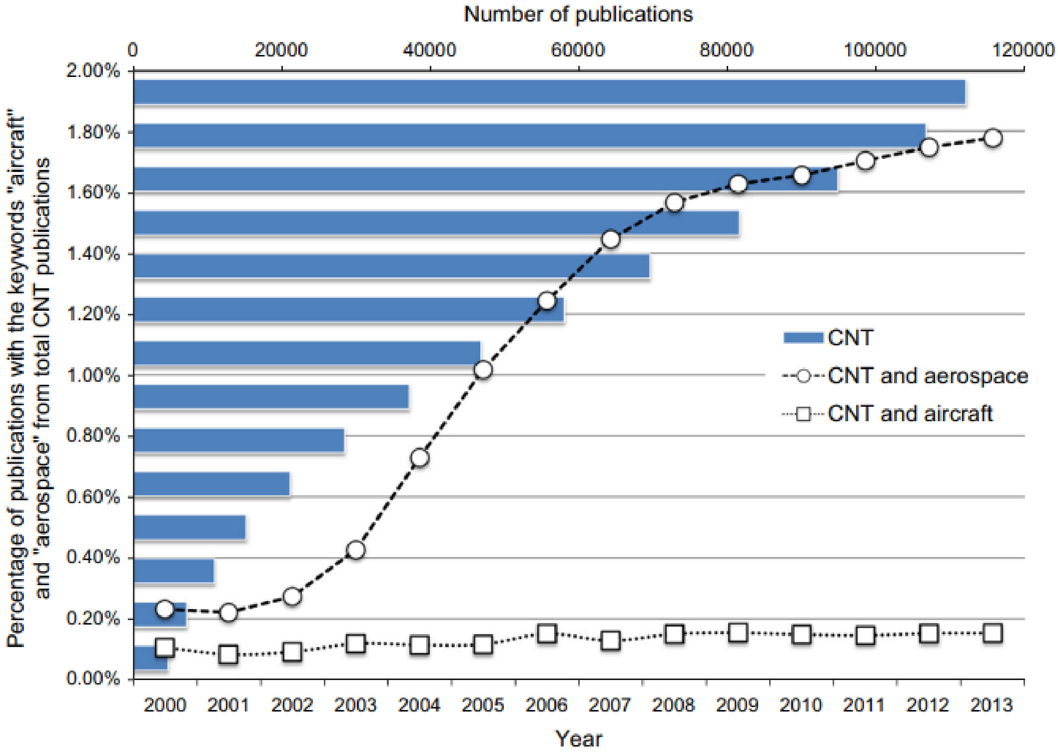

Fig 9 Graph showing No. of publications on CNTs with keywords as Aerospace and Aircraft

As you can observe there is a huge number of publications which indicates there has been a whole lot of research taking place related to CNTs. CNTs have shown remarkable properties which led to a whole lot of spectrum of advanced composite materials.

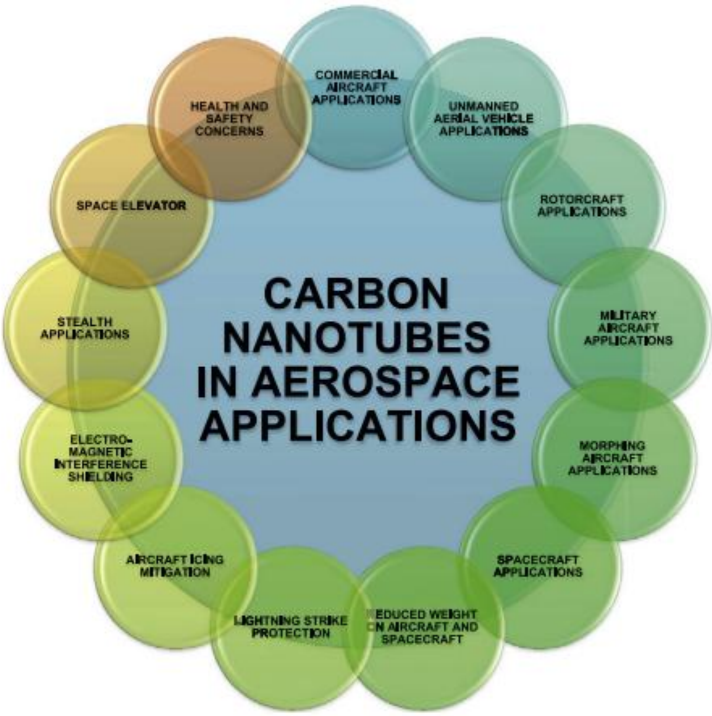

Fig 10 Different Applications of CNTs in Aerospace Industry

Composites

Composite materials are broadening the design frontiers of almost all engineering disciplines. In any event, the eye cannot perceive anything beyond the GRP1 yacht's exterior or overall racing performance, nor can it comprehend the intricacy of the shape of a composite rotor blade helicopter or of a contemporary tennis racket. However, this group of synthetic materials presents a chance for brand-new solutions to challenging engineering issues. In composites, materials are combined in this way to maximise our ability to use them while minimising to some extent the effects of their inadequacies. This optimization approach can free a fashion designer from the constraints associated with the choice and production of standard materials.

Materials with unusual properties are combined with or added to create composite materials. The unique components combine to give the composite its distinctive properties, yet inside the composite, you can tell the unique materials individually since they do not mix with one another. In nature, composites can be found. A piece of wood is a composite made of long cellulose fibers held together by lignin, a much weaker material. Although cellulose is also present in cotton and linen, it is the lignin that serves as the binding agent, making a piece of wood far more durable than a bundle of cotton fibers

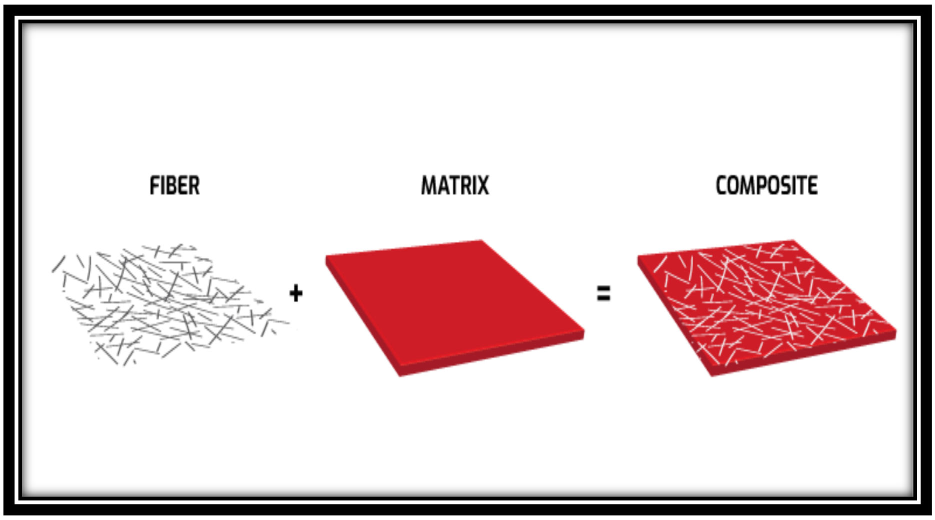

Fig 11 Formation of Composties

Metal matrix composites (MMC)

Metal matrix composites (MMC) are composite materials containing fibres or particles spread in a metallic matrix such as copper, aluminium, or steel, according to the field of materials science. The secondary phase is frequently a metal or ceramic (like alumina or silicon carbide). They are often divided into three categories based on the kind of reinforcement: short continuous fibres, short discontinuous fibres, or particles. MMC and cermets share several similarities despite the latter often containing less than 20% metal by volume. It is a hybrid composite if at least three different materials are included. Because it may have a substantially better strength-to-weight ratio, stiffness, and ductility than conventional materials, MMC is frequently utilised in demanding applications.

Matrix

The matrix is a continuous, monolithic material with embedded reinforcements. This implies that there is a channel through the matrix to any place in the material rather than two materials being sandwiched together. A lightweight metal, such as aluminium, magnesium, or titanium, serves as the matrix in structural applications and precisely supports the reinforcement. For high temperature applications, cobalt and cobalt-nickel alloy matrices are frequently used.

Reinforcement

The matrix contains the reinforcement. Reinforcement can be used to change physical attributes like wear resistance, coefficient of friction, and thermal conductivity in addition to strictly structural functions like strengthening connections. Both continuous and discontinuous reinforcement are possible. It is possible to process discontinuous MMC using extrusion, forging, or rolling since it is isotropic. Additionally, they are capable of being machined using traditional methods, however, PCD tools are frequently needed. Monofilament wires or fibres, such as silicon carbide or carbon fiber, are used in continuous armour. The orientation of the material determines its strength since the fibers are implanted in the matrix in certain orientations, resulting in an anisotropic structure.

Whiskers, or short fibres or particles, are used in discontinuous reinforcement. Aluminum oxide and silicon carbide are the most prevalent reinforcement materials in this group.

Stir Casting

The molten metal matrix is stirred during stir casting using a stirrer. Materials that can tolerate melting at temperatures higher than the matrix temperature are typically used to make stirrers. Castings are often stirred using graphite stirrers. The cylindrical rod and impeller make up the majority of the stirrer. The rod's other end is attached to the motor shaft, and its other end is attached to the impeller. A motor is often used to rotate the stirrer at different speeds while holding it in a vertical position. A mould is used to cast the resultant molten metal. Stir casting may be used to create composite materials with up to 30% reinforcement by volume. The segregation of reinforcing particles due to various process parameters and material qualities, which results in non-uniform metal distribution, is a significant issue with stir casting. Metal particle wetness, relative density, sedimentation velocity, and other process parameters are examples. Stirrer speed, stirrer angle, vortex cone, etc. all have an impact on how the particles are distributed inside the molten metal matrix. The matrix metal is first heated above its liquid temperature and made entirely molten in this procedure. It transitions into a semi-solid form when cooled to a temperature halfway between liquid and solid. The molten matrix is then mixed completely with the preheated reinforcing particles before being added and warmed to a fully liquid condition.

Carbon Nanotube- Reinforced Mg Composites

Recent observations show that diligent and critical research is being done to lighten materials utilised in a variety of applications without sacrificing strength. The materials utilised in automotive and aeronautical applications are the focus of these investigations. Magnesium is discovered to have a density that is even lower than that of titanium and aluminium. Magnesium's weak ductility and strength prevent it from being used in industrial settings. Due to its hexagonal close-packed (HCP) crystal structure, magnesium has poor characteristics. Magnesium has the aforementioned limitations, thus it must be improved in order to increase its usage in diverse applications. To do this, strengthening agents must be added through various processing methods.

To increase the strength and other characteristics of pure magnesium, nanoscale reinforcing agents such GNPs, Al2O3, SiC, and CNTs have been introduced into the magnesium matrix. CNTs are a desirable alternative to conventional Mg matrix-reinforced reinforcements due to their low density and appealing strength. CNTs may be used to create reinforced MMCs with a high degree of stiffness. This CNT is added to the created MMC to improve properties like high ultimate strength (30 GPa) and high elastic modulus.

However, compared to MMCs reinforced with Nanoplate, those made by reinforcing CNTs do not have as excellent of mechanical qualities (GNP). The inadequate improved bonding with the Mg matrix material is the cause of the lack of properties in CNT-engineered MMCs. Due to CNTs' low wetability on the Mg matrix material, there is weak bonding. Due to a number of deficient qualities, including: B. Low corrosion strength, poor formability, and poor mechanical properties, the use of magnesium and its alloys in a variety of applications is not recommended. The low standard electrode potential (-2.36 V), which interacts with other elements and causes Mg to corrode more quickly, is the cause of the low corrosion resistance of magnesium and its alloys. Since Fe, Cu, and Ni impurities frequently combine, the frequency of corrosion precipitation in Mg monoliths as a result is surprisingly unstable at the ppm level.

Experimental Procedure:

Ash Production:

The base product that was taken was sugarcane waste. The sugarcane waste was taken from a local vendor and has been put to dry outside for 4 days.

(a) (b)

Fig 12 (a) Sugarcane fibers (b) Sugarcane collected from vendor

The sugarcane waste was then weighed and then separated into fibers to put into a crucible. The weight of the sugarcane taken before the burning was 18gm.

Fig 13 Sugarcane fibers placed in Crucible.

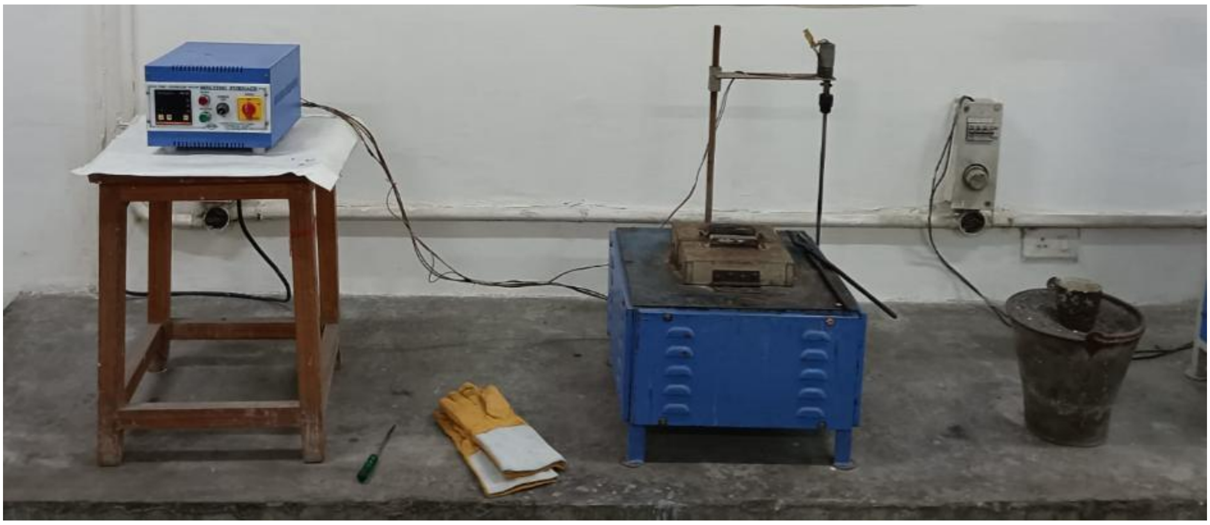

Fig 14 Melting Furnace Setup

A melting Furnace was the instrument that was used to break down the sugarcane fibres to ash. The melting furnace setup from fig 2 we can see a temperature controller and a furnace. The furnace can reach up to a temperature of 1000°C.

(a) (b)

Fig 15 Different parts of Melting furnace (a) temperature controllers (b) furnce

After the setup was started and the temperature controller was set to a particular temperature i.e., 400°C. The furnace will take some time to reach the desired temperature and then the crucible is placed in the furnace.

(a) (b)

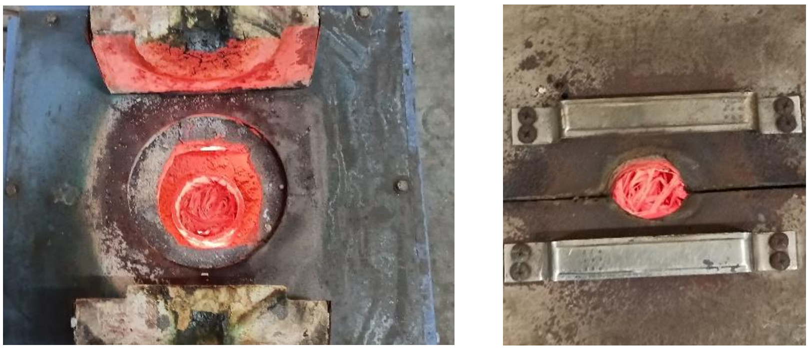

Fig 16 curcible with sugarcane fibers (a) without lid closed of furnace (b) with lid closed of the furnace.

The breakdown of the sugarcane fibers is done in 5 batches. Let the furnace reach the temperature that it was set to which is 400ºC in this experiment. The fibers were put in the crucible and then into the Melting Furnace maintaining a temperature of around 400ºC. All 5 batches of information is mentioned below in the table.

After the crucible which had burned down ash was kept away in the sand bucket to cool down. After the ash in the crucible cooled down, it was weighted each batch using Electronic weighing Balance machine.

(a) (b)

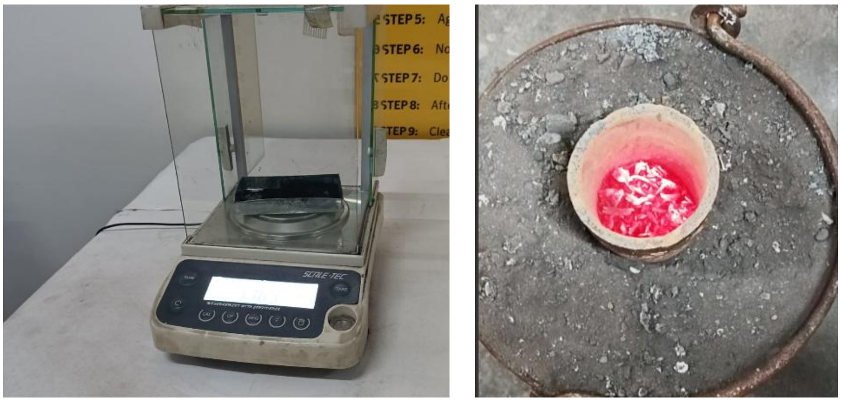

Fig 17 (a) Electronic weighing balance machine (b) Sand bucket

batches	Time (min: sec)	Initial weight of sugarcane waste(gm)	ash weight produced (gm)
1st batch	16:23	3	0.965
2nd Batch	15:08	7	2.188
3rd Batch	13:23	2.54	0.795
4th Batch	14:34	4.76	1.4902
5th Batch	11:18	0.7	0.1889

Table 4 Collective information of different Batches of burning ash

In the table above we can observe that the time for the burning mostly depends on the amount of sugarcane waste taken for that batch. The average time of the burning comes to be 13 minutes and a total of 18gm of sugarcane waste burned up to 5.6gm of ash.

(a) (b)

Fig 18(a) muffle furnace (b) compartments in muffle furnace storing the ash

This ash which is produced from the melting furnace consists of moisture, which is not desirable for the experiment. To remove the moister, ash that was collected before was transferred into the muffle furnace. A muffle furnace is used for heat treatment from which we can remove moisture. The 5.6gm of ash that was produced earlier was then put in a muffle furnace. It was put at the temperature of 400°C for 10 minutes. After the removal of the moisture, the weight of the ash that was acquired was 4.772gm.

After the ash preparation, we noted down all the chemical reactions that were to produce CNT. We have referred to different papers indicating the preparation of the CNT by chemical reactions.

Chemical Process from ash to CNT

Fig 15 Set-up for the chemical process

For the chemical process the man chemicals that we are going to use are Potassium Chlorate (($KClO_3$), ash that was produced before, Nitric Acid (HNO_3), and sulphuric acid (H_2SO_4). Collecting all the necessary beakers, measuring tubes, and also distilled water.

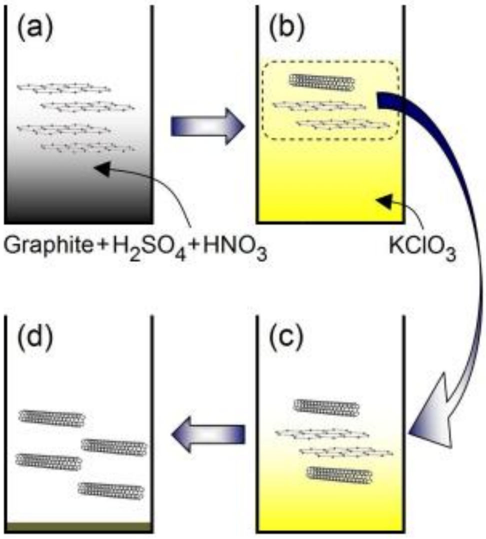

Fig 16 Chemical process for the production of CNTs

To start of the process 4.0m of ash is measured and taken into a beaker. Then add fuming nitric acid in the quantity of 20ml for 30 minutes. Add 40 ml of sulphuric acid for 30 minutes. These chemicals must be added slowly as they cause chemical reactions depicted in the figure above.

(a) (b)

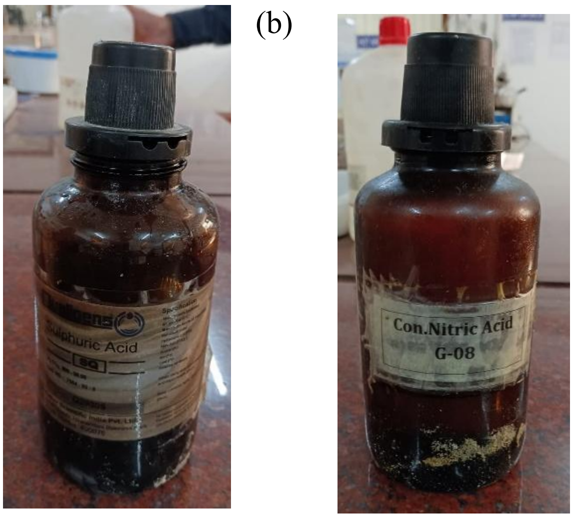

Fig 17 Chemicals used in the process (a) Sulphuric Acid (b) Nitric Acid

During these chemical reactions after the addition of Nitic acid, there are orange fumes are formed which indicates that the process is taking place correctly.

Fig 18 After addition of Fuming Nitric Acid

While adding these chemicals there is a lot of heat is produced as they are exothermic reactions. To cool down the mixture of sulphuric acid, nitric acid, and ash is put into an ice bath. This mixture is left to cool down in an ice bath till it attains 5^0C. A thermometer is put inside the mixture to monitor the temperature. After an approximation of 5 minutes, the mixture cools down.

Fig 19 Mixture cooling down before the addition of catalyst.

Now the catalyst Potassium Chlorate is taken in powder form is measured to get 20.0gm. As this reaction of catalyst and mixture is extremely exothermic, there was special care was taken. The special care was constant monitoring of temperature and placing the mixture in an ice bath during the whole process of adding the catalyst. Potassium Chlorate which acts as a catalyst has been added to the mixture slowly for 25 minutes under constant stirring. At the end of the addition of Potassium Chlorate, the mixture turns into a dark green semisolid. As a lot of fumes are released during this process, it recommended to take extra care while doing this process.

(a) 

(b) 

Fig 18 (a) Potassium chlorate measured to 20.0gm (b) Addition of Potassium chlorate to the mixture

This solution or semisolid mixture is then put in a hot air oven at around 70^0C for 26 hours. After the mixture was taken out of the furnace, it was left to open for 3 days. Then this was added to distilled water, and after 1 hour of stirring, the floating carbon was collected and dried.

Fig 19

(a) 

Hot air oven

(b) 

Sample mixture in the hot air over

This process from the addition of a catalyst is repeated multiple times until there is a significant amount of sample is produced. This sample is then sent for testing.

Sample testing

The sample that was produced from the two processes of breaking down of the sugarcane waste and then the chemical process to produce CNT. Now the sample is sent for testing under Scanning Electronic Microscope (SEM) and Transmission Electron Microscopy (TEM), and Raman Spectroscopy test. The testing piece was sent to infintelab, where they collect the sample and do the necessary tests and send the results to the inbox.

Stir Casting

For the production of Metal Matrix Composite, we have used Magnesium as the matrix and CNT as the reinforcement. This was done using Stir Casting. We have obtained Magnesium rods from the online store, Magnesium Rods (2 Pcs) 4 Inches X 1 Inch Diameter made of 99.8% Pure Magnesium Metal.

The setup is like the ash-burning process of the CNT. The melting furnace is used a same setup for Stir Casting. Where in the Crucible there is metal such as Magnesium and the reinforcement is CNTs is added in the melted Metal.

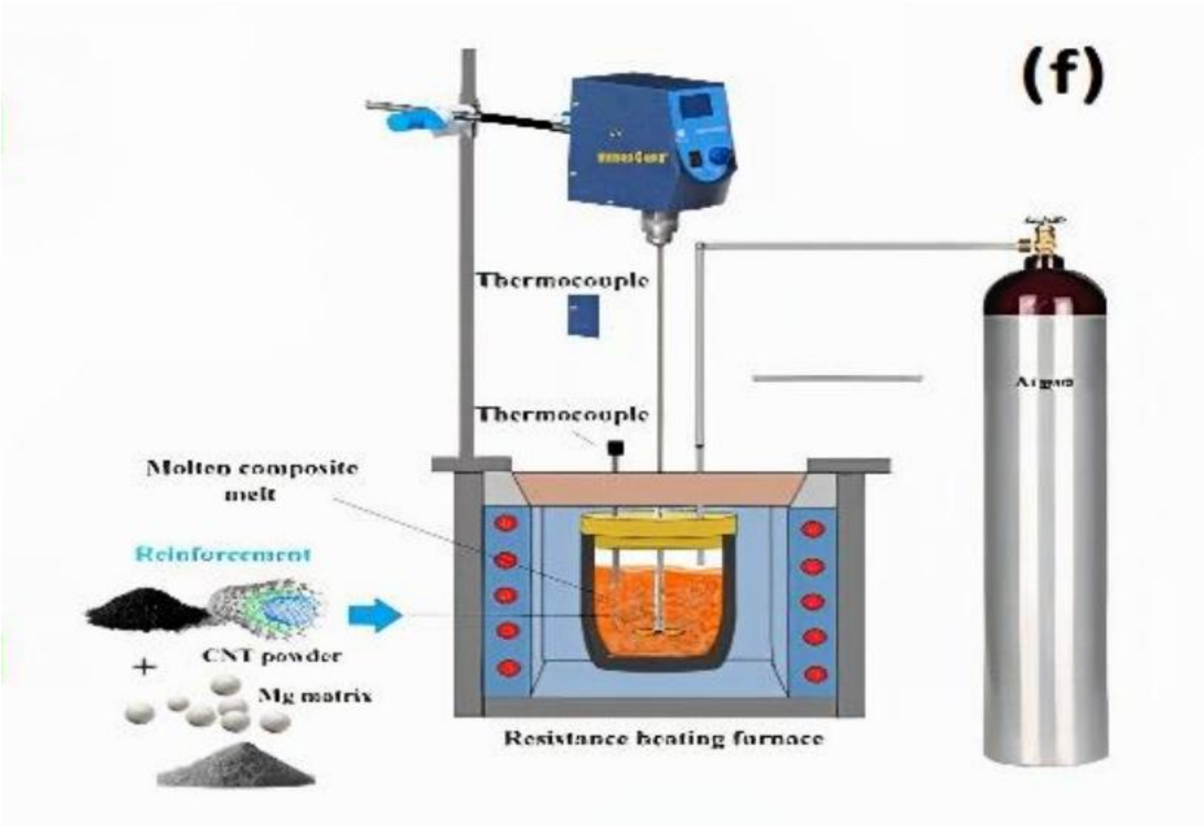

Fig 20 Stir casting process

In this process first, we melted the Mg rods we obtained; their melting point is 630^0C. After melting the Metal, we add the CNT as the reinforcements. We have added 0.5% of weight Mg of CNT. For this ratio, we have taken 1.25 gm of CNT and added it to the 250 gm of Mg. The stirring process was done for 7 mins for good results. After the Stir casting was done, the metal Matrix Composite is done in production.

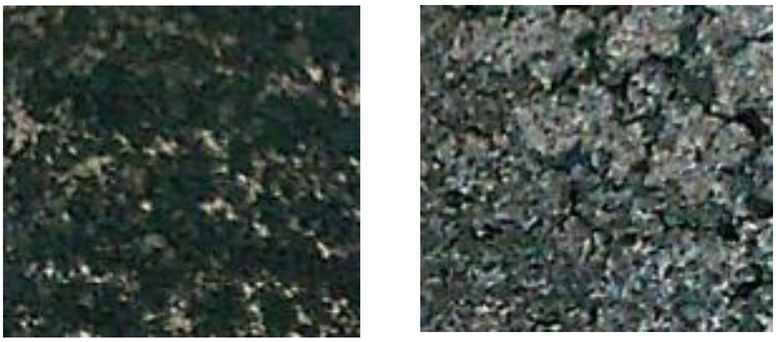

Fig 21 Composite under atomic microscope

Mechanical Testing

Mechanical testing such as compressive test and Tensile test of the composite is done using the Universal Testing Machine (UTM) from the Solid Mechanics lab of ASET in the Mechnical department.

Fig 22 UTM Universal Testing Machine

Results and Discussion

CNT analysis

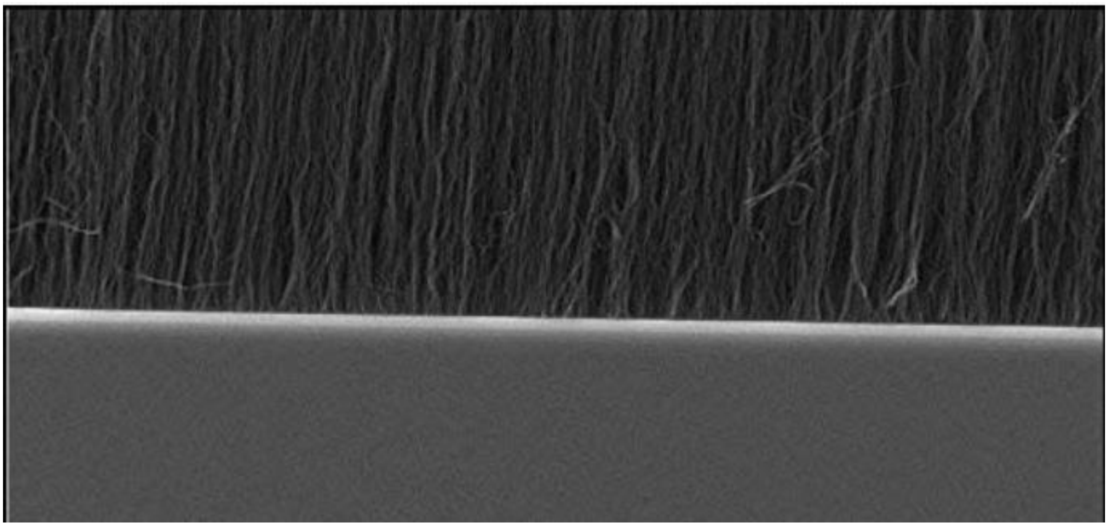

Fig 23 SEM results of the CNT Produced

From the SEM test results we have received; it can be concluded that the CNT that we produced was Multiwalled CNT. From this picture, it can also be observed that MWCNTs we produced were not much entangled.

Fig 24 TEM results of the CNT produced

From the TEM test results, it can observe the diameter of the MWCNTs. The inner diameter is observed to be between 10-20nm and the outer diameter to be between 30-50nm. From this, the surface area was found to be between 250-300 m^2/g and a length of 10 microns.

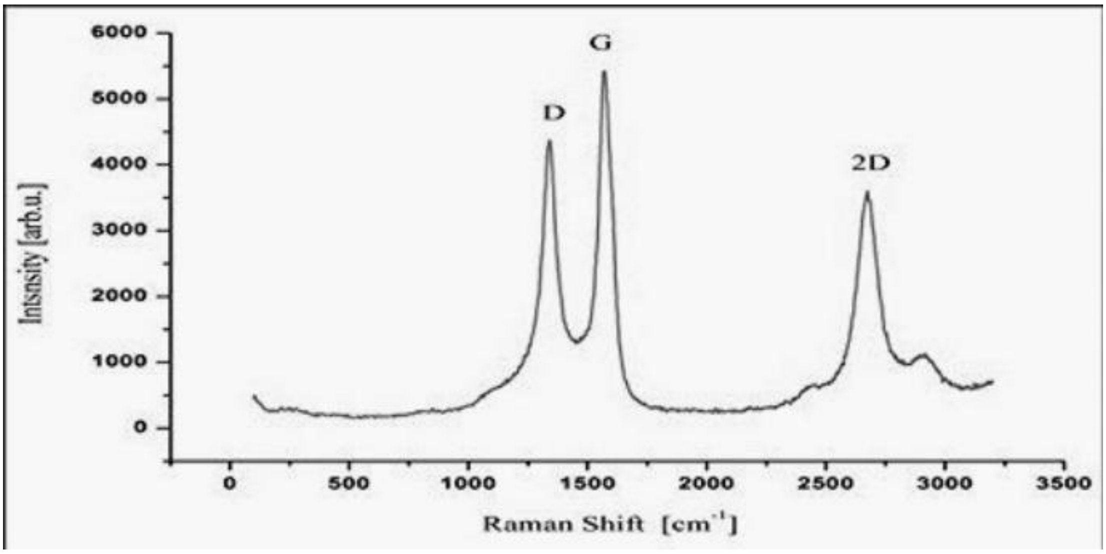

Fig 25 Raman Spectra Analysis

The graph from the Raman Spectra analysis there are main three peaks such as D peak which shows the disorders of the structure in the sample which is low and better and shows that there is very less disorderness. The next peak gives about the stretching of the C-C bond in the carbon-related structure and sp^2 carbon bonds. The distance of the peaks values of D and G peaks gives an idea of the diameters of the nanotube that is produced. And the relatively high peak of D gives the idea of an Amorphous structure being formed.

Mechanical properties

The composite that was prepared from the stir casting has been done testing, and the test results are here organized in table

Fig 26 Broken Mg-CNT composite after UTM testing

Material	Percent of CNT in the composite	Yield strength (MPa)	Compressive Strength (MPa)
Magnesium	Unreinforced	130.5	139.9
Magnesium-CNT composite	0.1	73.63	131.08
	0.2	87.06	144.25
	0.4	80.6	142.34
	0.5	77.1	148.82
	0.6	79.2	151.6

Table 5 Results from the UTM

From the result of the UTM, there is a decrease in the tensile strength which is decided from the Yield Strength (MPa). There is a decrease of 0.11% in the tensile strength of the composite compared to Magnesium. The compressive strength of the Composite has increased 25 % compared to the magnesium metal.

We have even added different amount of CNT in the composite to see which percentage can it yield more application. The ideal percentage for the addition of CNT in the composite is observed to be 0.5%

Conclusion

In Aerospace Industry, Metal Matrix Composites are playing a crucial role in redesigning Advanced Composite Materials. With this purpose of preparing advanced composite Materials in the industry, there is a huge surge of research in the CNT reinforced CNT. In this project, we have prepared the CNT from Agricultural waste, due to the high demand for Sustainable production and the easy access to the preparation of CNT for the easier axis of preparing the Advanced Composites. In this project, there was difficulty in the preparation of the CNT and to make it easily preparable. Sugarcane has high density of Carbon and burning it gives ash with high quantity of Carbon, allowing to do further chemical process and reactions for the easily available preparation of CNT. After the preparation of CNT, Metal Matrix Composite is prepared using Stir casting where the Matrix is the Magnesium and the Reinforcement is CNT that was prepared.

After the results analysis we were able to show that the Composite od CNT is more appilication used on the percentage of 0.5%.